

Data Flow Diagram

Date	04 July 2026
Team ID	
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	2 Marks

Data Flow Diagram

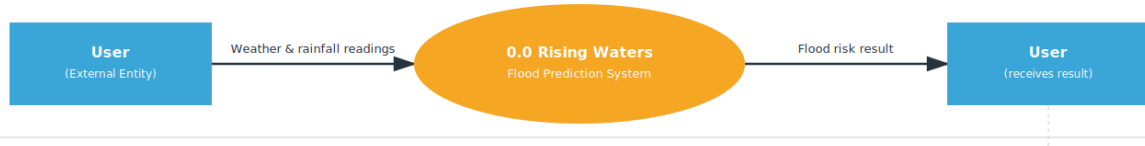
Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores.

DFD Symbol Legend

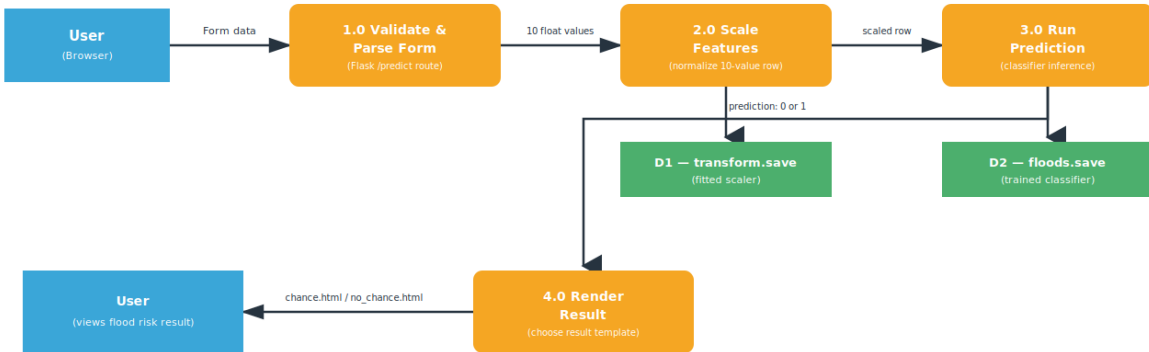
Symbol	Name	Description
Oval / Rounded shape	External Entity	A person or system outside the project that sends or receives data (e.g., the site visitor entering readings).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., 1.0 Validate Form, 2.0 Scale Features, 3.0 Run Prediction).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., floods.save, transform.save).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

DATA FLOW DIAGRAM — Rising Waters Flood Prediction System

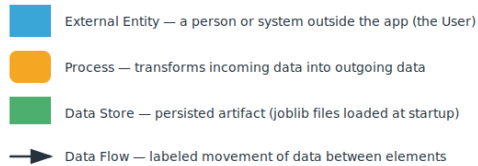
LEVEL 0 — Context Diagram



LEVEL 1 — Detailed Process Flow



Legend



Narrative

1. The User (external entity) submits the prediction form with 10 weather/rainfall values.
2. Process 1.0 validates and parses each field into floats inside the Flask /predict route.
3. Process 2.0 reads the fitted scaler from Data Store D1 (transform.save) and normalizes the row.
4. Process 3.0 reads the trained classifier from Data Store D2 (floods.save) and returns 0 or 1.
5. Process 4.0 renders chance.html or no_chance.html and returns the visual result to the User.

Rising Waters Data Flow

1. User (External Entity) submits the prediction form (Temp, Humidity, Cloud Cover, Annual, Jan-Feb, Mar-May, Jun-Sep, Oct-Dec, AvgJune, Sub) 2.0 Validate & Parse Form (Process) in the Flask /predict route converts fi each field to float 3.0 Scale Features (Process) reads the fitted scaler from Data Store: transform.save and fi normalizes the 10-value row 4.0 Run Prediction (Process) reads the trained classifier from Data Store: fi floods.save and returns 0 or 1 5.0 Render Result (Process) selects chance.html or no_chance.html and returns it to the User (External Entity).